

NEWTON, IA, USA

WMO: 725464

Lat: 41.674N

Lon: 93.022W

Elev: 291

StdP: 97.88

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04977

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.8	-17.7	-25.1	0.4	-20.2	-22.4	0.5	-17.2	13.5	-8.7	12.0	-6.4	3.6	320	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	32.8	25.2	32.0	24.9	30.1	23.6	27.0	31.2	25.8	29.9	24.8	28.9	4.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	22.2	30.3	24.8	20.6	29.1	23.0	18.4	27.3	87.6	31.4	81.6	29.5	77.2	29.3	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	10.0	8.8	-25.7	34.8	3.2	1.6	-28.0	35.9	-29.9	36.9	-31.7	37.8	-34.1	39.0		
			-24.7	28.7	3.8	1.4	-27.4	29.7	-29.6	30.5	-31.8	31.3	-34.5	32.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-5.2	-3.7	3.3	10.3	16.4	21.7	23.8	22.3	18.4	11.4	3.6	-2.8	(6)
(7)		DBStd	11.56	6.86	7.09	6.84	5.49	4.62	3.38	2.99	3.16	4.44	5.39	5.84	6.29	(7)
(8)		HDD10.0	1797	472	384	227	62	5	0	0	1	47	203	396	(8)	
(9)		HDD18.3	3538	730	616	467	246	93	9	1	53	222	441	654	(9)	
(10)		CDD10.0	1812	0	1	19	70	203	351	429	381	254	92	12	1	(10)
(11)		CDD18.3	512	0	0	1	4	33	110	172	128	56	8	0	0	(11)
(12)		CDH23.3	4673	0	0	10	75	319	982	1631	1045	522	87	3	0	(12)
(13)		CDH26.7	1587	0	0	1	17	90	333	617	336	172	22	0	0	(13)
(14)	Wind	WSAvg	4.2	4.7	4.6	4.9	5.2	4.7	3.8	3.0	2.8	3.4	4.2	4.5	4.4	(14)
(15)	Precipitation	PrecAvg	922	23	31	51	96	121	143	119	111	105	58	47	27	(15)
(16)		PrecMax	1402	52	72	132	227	236	339	285	286	190	147	134	91	(16)
(17)		PrecMin	630	9	5	1	35	18	26	14	14	46	12	0	0	(17)
(18)		PrecStd	186	10	17	38	47	48	72	74	74	40	39	34	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	17.0	24.8	29.7	32.4	33.9	35.7	33.9	33.0	29.8	22.7	15.2	(19)
(20)			MCWB	6.9	10.6	16.7	18.5	23.0	24.7	27.5	25.0	22.7	19.8	14.7	12.0	(20)
(21)		2%	DB	8.7	12.4	20.7	25.2	29.0	32.1	32.9	32.1	31.1	26.0	18.0	11.2	(21)
(22)			MCWB	5.5	8.5	14.3	16.2	20.7	24.7	26.4	25.3	22.8	17.8	12.5	8.2	(22)
(23)		5%	DB	6.0	8.0	17.3	22.5	27.2	30.2	32.1	30.1	28.7	22.7	15.1	7.8	(23)
(24)			MCWB	3.4	5.4	12.7	14.6	18.7	23.0	25.9	24.3	21.6	16.6	10.4	5.5	(24)
(25)		10%	DB	2.9	5.9	13.6	20.0	24.8	28.6	30.1	28.0	27.0	20.1	12.4	5.2	(25)
(26)			MCWB	1.4	3.7	9.1	13.1	18.0	22.3	24.7	23.0	20.1	15.3	8.6	3.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	11.8	18.0	20.0	24.9	27.6	29.3	28.3	25.7	21.7	15.9	12.9	(27)
(28)			MCDWB	11.3	16.0	22.6	26.2	29.9	32.0	34.2	32.2	30.8	26.2	20.5	15.3	(28)
(29)		2%	WB	5.6	8.7	15.6	17.6	22.1	25.7	27.6	26.4	24.0	19.6	13.6	9.0	(29)
(30)			MCDWB	8.7	12.3	19.2	23.3	27.3	29.9	32.1	30.2	28.7	23.6	18.0	10.2	(30)
(31)		5%	WB	3.5	5.9	12.6	15.7	20.5	24.4	26.4	25.1	22.7	17.9	11.1	5.6	(31)
(32)			MCDWB	5.7	8.5	16.2	21.1	25.0	28.8	30.3	29.0	26.9	21.1	14.6	8.0	(32)
(33)		10%	WB	1.7	3.7	9.6	14.0	19.1	23.2	25.3	24.0	21.4	15.8	8.8	3.5	(33)
(34)			MCDWB	3.3	5.6	13.6	18.7	23.0	27.3	29.3	27.5	25.1	19.6	12.3	5.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.8	9.0	10.1	11.7	10.9	10.5	10.3	12.3	11.1	9.8	8.3	(35)	
(36)			MCDBR	12.4	12.4	15.0	16.4	13.8	12.2	11.7	11.6	14.0	14.4	14.4	11.4	(36)
(37)			MCWBR	9.5	9.0	9.6	9.3	7.0	6.8	7.2	7.1	7.4	8.0	9.4	8.9	(37)
(38)		5% WB	MCDBR	11.9	12.0	13.6	13.9	11.7	11.0	10.6	10.2	11.8	11.9	13.2	10.7	(38)
(39)			MCWBR	9.2	9.1	9.5	9.2	7.1	7.4	7.5	7.1	7.2	8.1	9.7	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.288	0.310	0.331	0.380	0.409	0.420	0.437	0.417	0.385	0.341	0.300	0.288	(40)	
(41)		taud	2.408	2.353	2.392	2.274	2.265	2.279	2.216	2.289	2.331	2.448	2.494	2.429	(41)	
(42)		Ebn at Noon	865	896	918	889	868	856	836	844	847	851	848	833	(42)	
(43)		Edh at Noon	79	98	106	129	133	132	139	126	112	89	73	71	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.91	2.92	3.80	4.73	5.58	6.18	6.26	5.43	4.58	3.05	2.07	1.58	(44)	
(45)		RadStd	0.18	0.20	0.36	0.36	0.42	0.60	0.45	0.34	0.34	0.36	0.18	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page